CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 7: **Work, Energy and Simple Machines** | Quick revision, one page

Work

- $W = F\,s$ for force along displacement; SI unit joule, $1\ \text{J} = 1\ \text{N m}$.
- Work is zero if the displacement is zero or the force is perpendicular to the displacement; negative if the force opposes the displacement (e.g. friction).
- Work-energy theorem: work done by the net force = change in kinetic energy.

Energy

- Kinetic energy $KE = \frac{1}{2}mv^2$; potential energy (gravitational) $PE = mgh$.
- Law of conservation of energy: energy can neither be created nor destroyed, only changed from one form to another.
- Mechanical energy $= KE + PE$ is constant for a freely moving body (no friction): $\frac{1}{2}mv^2 + mgh = \text{constant}$; speed after falling through h from rest $v = \sqrt{2gh}$.

Power

- $P = \dfrac{W}{t} = \dfrac{E}{t}$; SI unit watt, $1\ \text{W} = 1\ \text{J/s}$; $1\ \text{kW} = 1000\ \text{W}$.
- Commercial unit of energy: $1\ \text{kWh} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J}$.
- Energy used (kWh) = power (kW) $\times$ time (h).

Simple machines

- Mechanical advantage $MA = \dfrac{\text{load}}{\text{effort}}$ (no unit). Lever: $MA = \dfrac{\text{effort arm}}{\text{load arm}}$; principle of moments: effort $\times$ effort arm = load $\times$ load arm.
- Fixed pulley $MA = 1$ (changes direction only); movable pulley $MA = 2$; inclined plane $MA = \dfrac{\text{length of ramp}}{\text{height}}$.
- A machine allows a smaller effort to move a larger load but does not reduce the work: ideal $W_{\text{in}} = W_{\text{out}}$.

Efficiency

- Efficiency $= \dfrac{\text{useful work (output)}}{\text{work put in (input)}} \times 100\%$; always $< 100\%$ for real machines because of friction.

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